PROJECT INTERVIEW PREPARATION

Etihad Travel Assistant

100 interview questions

A project-specific question bank covering architecture, policy RAG, hybrid search, AviationStack MCP, LangGraph routing, Redis caching, security, observability, and testing.

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Based on the current local project implementation

Questions only - use them for mock interviews, design reviews, and self-assessment.

Project and architecture

Purpose, system boundaries, and a request's path through the application.

001 What user problem does the Etihad Travel Assistant solve, and which requests are deliberately outside its scope?

002 How does a request move from the Streamlit chat input to the FastAPI response and back to the user?

003 Why did you choose FastAPI for the backend and Streamlit for the user interface in this local MVP?

004 How does the application distinguish a policy question from a live flight-information request?

005 What responsibilities do LangChain, LangGraph, and LangSmith each have in this project?

006 Which components own policy data, live flight data, conversation history, and reusable answers?

007 What is the complete path of a first-time baggage-policy question, including retrieval, refinement, citations, and caching?

008 What is the complete path of a flight-status question when the user initially omits the departure date?

009 Which parts of this project are deterministic, and which parts depend on a language model or an external API?

010 What trade-offs did you accept by building a local MVP rather than a production deployment?

Policy ingestion and sources

Corpus creation, provenance, policy freshness, and source quality.

011	How is the Etihad Hub policy library collected, and what completeness checks prevent a partial scrape?
012	Why does the project preserve original HTML, normalized text, source links, and scrape metadata?
013	How are policy documents divided into retrieval chunks, and what can go wrong if chunks are too large or too small?
014	How does the system keep supplemental official Economy baggage information separate from the Hub snapshot?
015	What does the corpus digest represent, and how is it useful when the underlying policy documents change?
016	How would you detect that a scraped policy is outdated, superseded, or effective only on a future date?
017	What would you do if two official Etihad pages appeared to disagree on the same allowance?
018	How do you preserve section paths and citations so a retrieved answer can be traced to its source?
019	What tests would confirm that a refresh has not silently dropped a policy group or important table?
020	How would you handle a page redesign that changes the scraper's expected HTML structure?

Hybrid search and Chroma

BM25, embeddings, vector retrieval, fusion, and relevance evaluation.

021 How does BM25 rank policy chunks, and which types of exact-match questions does it handle well?

022 How are OpenAI embeddings and persistent Chroma collections used for semantic policy retrieval?

023 Why is combining BM25 and semantic search preferable to using either method alone for this corpus?

024 What was the effect of the original BM25-first early return, and how does the current retriever avoid it?

025 How are lexical and vector candidates combined, and what is the purpose of rank fusion?

026 When might a semantic search return a related but incorrect policy, and how should that result be filtered?

027 How would you tune the number of retrieved chunks without overwhelming the answer-refinement step?

028 How does Chroma reuse an existing policy index after an application restart, and when must it re-embed?

029 What should happen if Chroma or the embedding API fails while BM25 still has a useful match?

030 Which evaluation questions would you use to compare BM25-only, semantic-only, and hybrid retrieval?

RAG and answer grounding

Evidence selection, OpenAI refinement, citations, and honest uncertainty.

031 After retrieval, how does the project select question-relevant passages instead of sending whole documents to the LLM?

032 Why does the application use an LLM to refine a retrieved policy answer rather than display every retrieved passage?

033 What does 'grounded answer' mean in this implementation, and how are claims connected to citations?

034 What should the assistant say when retrieved excerpts do not actually establish the requested policy detail?

035 How does the answer prompt prevent invented baggage weights, dates, fees, or refund exceptions?

036 How does the system answer an Economy baggage question when the traveler has not specified the fare type?

037 Why might a broad Platinum-benefits question need multiple source excerpts rather than one chunk?

038 How should the assistant handle a general onboard-food question without inventing a flight-specific menu?

039 How could malicious instructions embedded in a scraped policy page affect RAG, and what defenses exist?

040 What additional checks would you add before treating an LLM-generated policy answer as safe to cache?

LangGraph routing and memory

Intent classification, clarification turns, state, and conversation isolation.

041 What are the major LangGraph branches for policy, flight, clarification, off-topic, and fallback responses?

042 How is the route selected before policy retrieval or a flight tool is invoked?

043 What state must be retained when the assistant asks for a missing airport or flight date?

044 Why should a reply such as 'I said Abu Dhabi' continue the pending flight request instead of starting a new topic?

045 How is recent conversation history scoped to the authenticated user and conversation?

046 Why is chat history stored separately from the policy vector index?

047 When should a context-dependent follow-up bypass the shared policy-answer cache?

048 What happens when a policy question has no useful retrieval evidence but is not clearly off-topic?

049 Which routing rules should stay deterministic even if an LLM could classify intent?

050 How would you debug a user question that was incorrectly routed to RAG instead of the flight MCP tool?

AviationStack and MCP

Live provider calls, normalized tool results, and no-fabrication behavior.

051	Why does live flight information come from AviationStack through MCP instead of from the policy corpus?
052	Which flight tools are exposed by the MCP layer, and what is each tool responsible for?
053	How is the first flight from Delhi to Abu Dhabi chosen from provider records for a given date?
054	How are city names, airport aliases, and IATA codes converted into tool parameters?
055	How are AviationStack's recognized flight-status values preserved in the user-facing answer?
056	Why must a future scheduled flight not be described as live operational status?
057	How does an airport-arrivals request reach the arrival-board tool rather than policy RAG?
058	What should the assistant report when AviationStack returns an empty result, an API error, or a plan-restricted endpoint?
059	Why is synthetic flight data prohibited in this project, including during provider outages?
060	How would you test date filtering, route matching, sorting, and MCP result normalization without making live API calls?

Redis, SQLite, and cache design

Shared answer reuse, persistence boundaries, invalidation, and privacy.

061 What information is included in a policy-answer cache key, and why is the raw question normalized?

062 Which policy questions are intentionally excluded from the shared Redis cache, and why?

063 How does a Redis cache hit change the number of embedding and chat-model calls?

064 What happens when Redis is unavailable, and why does the project avoid a SQLite cache fallback?

065 What is the purpose of a cache time-to-live, and how would you choose one for changeable airline policies?

066 Why does a remote Redis Cloud connection require TLS in the default configuration?

067 How do a corpus change, model change, or prompt/retrieval version change invalidate old answers?

068 Which data belongs in SQLite, which belongs in Chroma, and which belongs in Redis?

069 How would you prevent personally identifiable booking information from being shared through a cross-user cache?

070 How might you prevent many simultaneous users from triggering duplicate LLM calls for the same uncached question?

FastAPI, authentication, and RBAC

HTTP contracts, role enforcement, validation, and secret handling.

071	How does a user obtain a JWT, and how is that token checked on protected API endpoints?
072	How are the user and admin roles enforced on backend routes rather than only hidden in the UI?
073	Which API endpoints are public, user-accessible, or admin-only in the current application?
074	Why must the backend validate conversation ownership on every history or feedback operation?
075	How are passwords stored, and why would plaintext passwords be unacceptable?
076	What input validation is needed for chat messages, airport codes, dates, and dashboard limits?
077	Where should OpenAI, AviationStack, LangSmith, Redis, and JWT secrets be configured?
078	How would you prevent an ordinary user from requesting the admin dashboard directly via HTTP?
079	What should the API return when a flight provider is unavailable versus when a policy answer is not found?
080	What would you change in authentication and deployment before exposing this MVP publicly?

UI and observability

Streamlit experience, usage metrics, audit data, and troubleshooting.

081	How does the Streamlit interface present answers, citations, clarifying questions, and feedback?
082	Which metrics does the admin dashboard expose for LLM usage, cache activity, and application behavior?
083	What is the difference between a LangSmith trace, a SQLite LLM-usage row, and a JSONL audit event?
084	How would you calculate the cache hit rate and the number of avoided chat-model calls?
085	Which token counts, latency values, and error fields are useful when diagnosing model cost or failures?
086	Why should raw user messages be minimized or omitted from operational audit logs?
087	How would you trace a wrong answer back through routing, retrieval, source passages, and LLM refinement?
088	How can the UI make it obvious whether an answer comes from a policy source or a live flight provider?
089	What should the health endpoint report, and what should it avoid disclosing to anonymous callers?
090	How would you assess whether the redesigned Etihad-style UI improves usability without confusing users about official status?

Guardrails, tests, and next steps

Safety, regression coverage, evaluation, and considered improvements.

091 How does the assistant politely reject off-topic requests without rejecting valid questions about meals or loyalty benefits?

092 How does the project avoid promising a refund, waiver, or compensation that written policy does not guarantee?

093 What regression test proves semantic retrieval still runs when BM25 has no match?

094 What tests cover flight clarification across turns, especially a missing date or destination airport?

095 How can fake embeddings, fake Redis, and stubbed AviationStack responses keep tests deterministic?

096 Which test cases would catch a cached answer whose citations are no longer present in the current corpus?

097 How would you measure retrieval relevance and answer faithfulness using a project-specific evaluation set?

098 What are the most important remaining limitations of the current local MVP?

099 Which improvement would you prioritize next: better source freshness, retrieval evaluation, flight coverage, or stronger security - and why?

100 In a two-minute interview answer, how would you explain this project's impact and its main engineering trade-offs?